Kousumi Mondal

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PROFESSIONAL SUMMARY

Final-year Computer Science student and AI Engineer specializing in Machine Learning, Generative AI, and Intelligent Systems. Experienced in architecting end-to-end ML pipelines, production-grade RAG applications, and scalable backend services, building data-driven AI solutions across NLP, Agentic AI, and semantic retrieval with an emphasis on performance, scalability, and reliability.

EDUCATION

Vellore Institute of Technology — Bhopal, India
B.Tech Computer Science & Engineering – CGPA: 8.07 — *2023–2028*

Ideal Public School — Howrah, India
ISC Class XII Science – 71.08% — *2021–2023*

Ideal Public School — Howrah, India
ICSE Class X – 87.4% — *2013–2021*

PROJECTS

DRUNet Image Denoiser | *Python, PyTorch, OpenCV, Transformers* — GitHub | Live

– Developed a DRUNet-based image denoising model on the DIV2K dataset, achieving high-quality image restoration through deep residual learning techniques.
– Built an end-to-end training pipeline incorporating realistic noise injection, data augmentation, and optimization strategies for robust model generalization.
– Evaluated performance using PSNR and SSIM metrics and optimized inference workflows for efficient processing of high-resolution images.

MedInsight – Medical Intelligence System | *Python, LangChain, RAG, ChromaDB* — GitHub | Live

– Built a medical AI platform combining LangChain, Groq-hosted LLMs, and ChromaDB to enable retrieval-augmented clinical question answering over biomedical literature.
– Designed semantic retrieval pipelines for research paper processing, vector indexing, and context-aware response generation using biomedical data sources.
– Developed a FastAPI backend and JavaScript frontend, while leveraging Biopython for structured parsing and extraction of biological research information.

LexTrace – Legal Document Intelligence System | *FastAPI, LangChain, Groq, MongoDB* — Live

– Engineered a full-stack RAG pipeline for legal document ingestion and querying, leveraging PyMuPDF and Tesseract OCR to process both scanned and digital legal documents.
– Implemented semantic retrieval using Sentence Transformers and MongoDB vector search, enabling context-aware retrieval across large legal corpora with reduced query latency.
– Integrated a Groq-hosted LLM through LangChain for citation-aware legal question answering and developed a responsive TypeScript frontend for seamless user interaction.

TECHNICAL SKILLS

Production biased: Python, SQL, ETL, RAG, Agentic AI, Automation, Git, Java
Proficient: PyTorch, Scikit-learn, Pandas, Numpy, LangChain, React, DSA
Implementation: LangChain, Sentence Transformers, Vector Search, RAG Pipelines, MongoDB, ChromaDB
Familiar: Docker, AWS, Data Scraping, Cloud Fundamentals
Fundamentals: Data Structures, Algorithms, Databases

ACHIEVEMENTS

Meta X PyTorch OpenEnv Hackathon – Finalist — Vellore Institute of Technology, Bhopal
Selected among 800 finalist teams from 31,000+ registered participants nationwide. — *2026*